

Homework 3 Solutions

Question 1: Data Selection for Supervised Learning

(a) Which clusters do the data points selected by the Forgetting Score Algorithm belong to?

Solution:

- The **Forgetting Score Algorithm** generally selects data points that are misclassified or challenging to classify, i.e., points close to the decision boundary.
- In Figure 1, the data points closest to the boundary are primarily from **Cluster 1** and **Cluster 2**.
- Thus, we can deduce that the Forgetting Score Algorithm would likely select data points from **Cluster 1 and Cluster 2**.

Answer: E. Cluster 1, Cluster 2

(b) Which clusters do the data points selected by CRAIG belong to?

Solution:

- The **CRAIG Algorithm** aims to create a representative subset that covers the data diversity, potentially including points from each cluster.
- Since CRAIG tries to balance representation, it would likely select points from **all clusters (1, 2, 3, and 4)** to ensure diversity in the selected subset.

Answer: O. Cluster 1, Cluster 2, Cluster 3, Cluster 4

(c) Is the subset selected by the Forgetting Score Algorithm likely to achieve the same test accuracy as training on the full noisy training data?

Solution:

- The Forgetting Score Algorithm selects challenging or misclassified examples, which may not cover the entire feature space due to its focus on hard-to-classify points.

- This selection could lead to a less diverse subset, potentially limiting generalization.
- Therefore, training on this subset might not achieve the same test accuracy as training on the full noisy dataset, which has a more complete representation of the data.

Answer: B. No

(d) Is the subset selected by CRAIG likely to achieve similar test accuracy to training on the full noisy training data?

Solution:

- Since CRAIG selects a representative subset across all clusters, it should retain more diversity and provide better coverage of the feature space.
- This representative nature suggests that training on the CRAIG subset could yield similar test accuracy to training on the full dataset, even with noise.

Answer: A. Yes

Summary of the Solution Methodology

- **Cluster Selection:** We analyzed the tendencies of the algorithms (Forgetting Score focusing on difficult points near the boundary, CRAIG ensuring diversity).
- **Test Accuracy:** We reasoned that a diverse selection (CRAIG) would perform similarly to the full noisy dataset, while a subset with only difficult examples (Forgetting Score) might not generalize as well.

By considering each algorithm's behavior and the provided data distribution, we derived these answers.